

Universidad Nacional de San Agustín de Arequipa
School of Systems Engineering · Software Quality · Partial Work 6

Evaluating the quality of a malaria surveillance dashboard with ISO/IEC 25010

Quality model · GQM metrics · evaluation process & data-collection format · control board

Juan Manuel Cari Quispe · Daniel Wilston Chura Monroy · Ower Frank Lopez Arela
Alejandro Josue Phocco Tapia · Mauricio Eduardo Zegarra Puma
Lecturer: Mg. Jesús Martín Silva Fernández · Arequipa, Peru · October 2026

Agenda and team responsibilities

#	Topic	Presenter
1	Context, official data and the product (web dashboard + quality board); Point 1 — characteristics	Juan Manuel Cari Quispe
2	Point 2 — GQM and metrics: functional suitability, performance, compatibility, flexibility, maintainability	Daniel Wilston Chura Monroy
3	Point 2 — Metrics: interaction capability, reliability, security, safety	Ower Frank Lopez Arela
4	Point 3 — Evaluation process, data-collection format, failure classification	Alejandro Josue Phocco Tapia
5	Point 4 — Control board and evolution of the runs	Mauricio Eduardo Zegarra Puma
6	Results across the runs	Daniel Wilston Chura Monroy
7	Incident log	Ower Frank Lopez Arela
8	Lessons learned	Alejandro Josue Phocco Tapia
9	Conclusions	Mauricio Eduardo Zegarra Puma

Context: malaria in Peru and the official data

Source: MINSA – CDC Peru, National Open Data Platform (downloaded 2026-10-26)

1,243,299
notified cases 2000–2025

33,516
cases in 2025

15.4 %
P. falciparum share in 2025

99 %
of 2025 cases in Loreto

Official file	Period	Granularity	Rows	Invalid
datos_abiertos_vigilancia_malaria_2000_2008.csv	2000-2008	week × district aggregate	106,355	0
datos_abiertos_vigilancia_malaria_2009_2025.csv	2009-2025	nominal (one case per row)	630,657	40

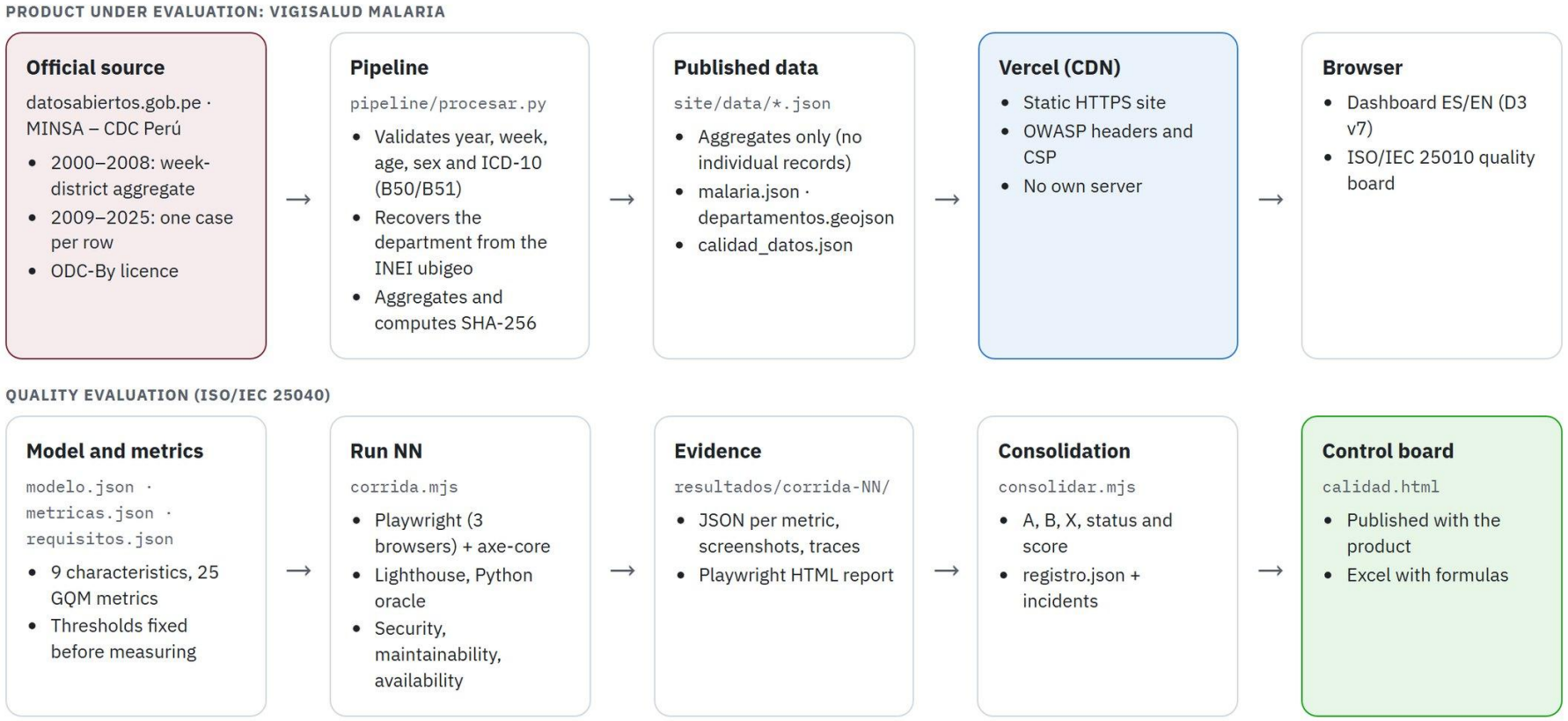
- **Validation:** year, epidemiological week, age, sex and ICD-10 code (B50 P. falciparum, B51 P. vivax) are checked on every row.
- **Recovery, not invention:** 5,712 cases without a department name were recovered from the INEI ubigeo code; the letter Ñ, delivered as U+FFFD in the official files, was restored in 14 names (figures unchanged).
- **Privacy by design:** only aggregates are published, each source file is fingerprinted with SHA-256.

The product we evaluate: VigiSalud Malaria

Public web dashboard + ISO/IEC 25010 quality board · https://vigisalud-malaria.vercel.app

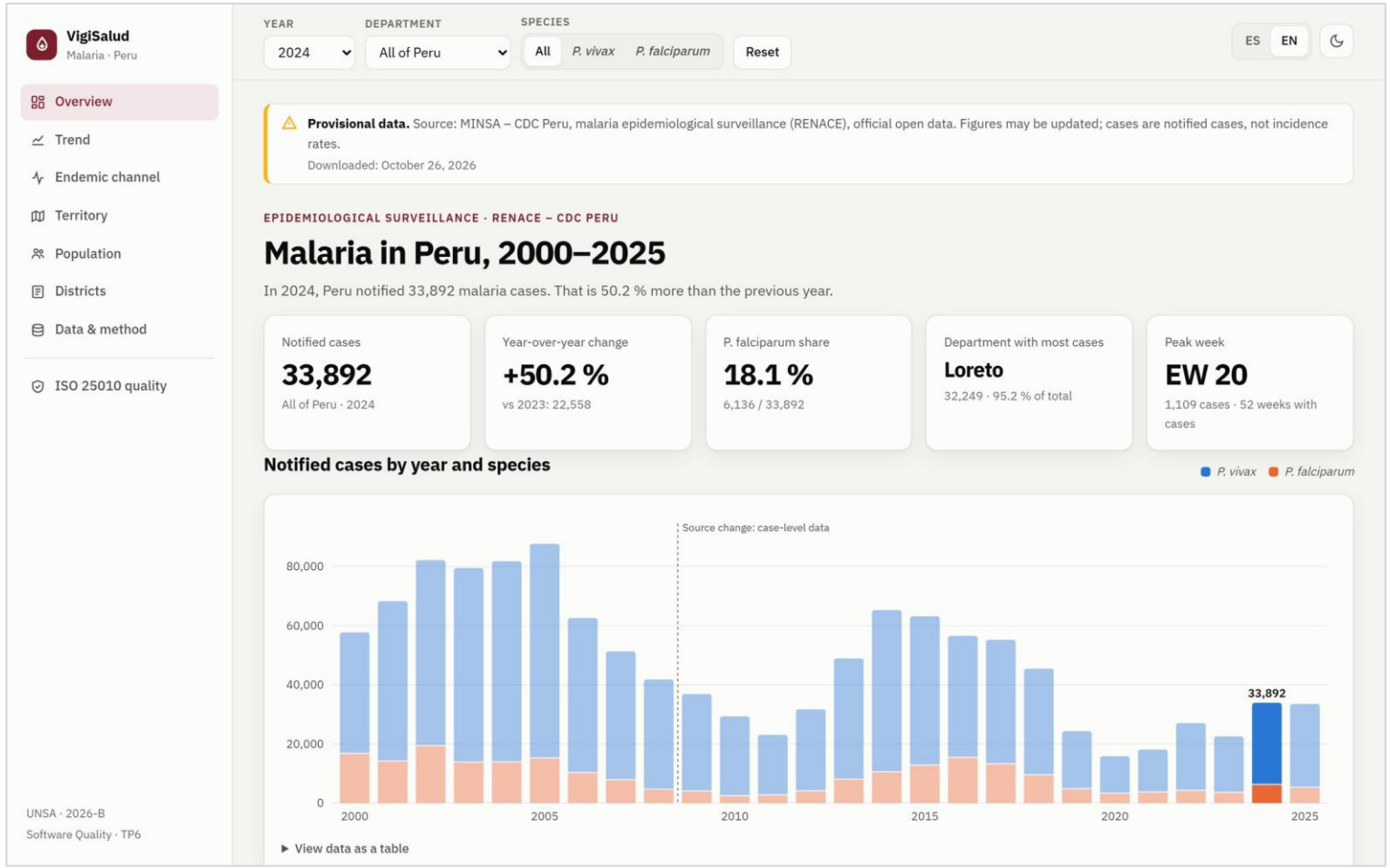
Product and evaluation architecture

Official data is processed once; the browser only receives aggregates. The evaluation measures the published version and feeds the control board.



The web dashboard (1/2): what the user sees

Three global filters — year, department, species — drive every view and are shared in the URL

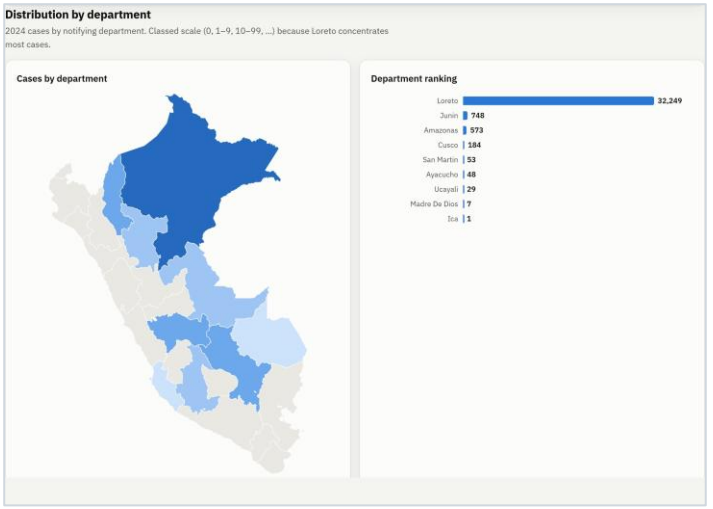


- **Navigation:** seven sections plus the ISO 25010 quality board.
- **Filters:** year, department and species; shared in the URL.
- **Provisional-data notice:** source, limits and download date.
- **Overview:** 5 KPIs and a one-sentence trend summary.
- **Trend:** annual cases by species; dashed line marks the 2009 change of source.
- **ES / EN and light / dark:** remembered per user.

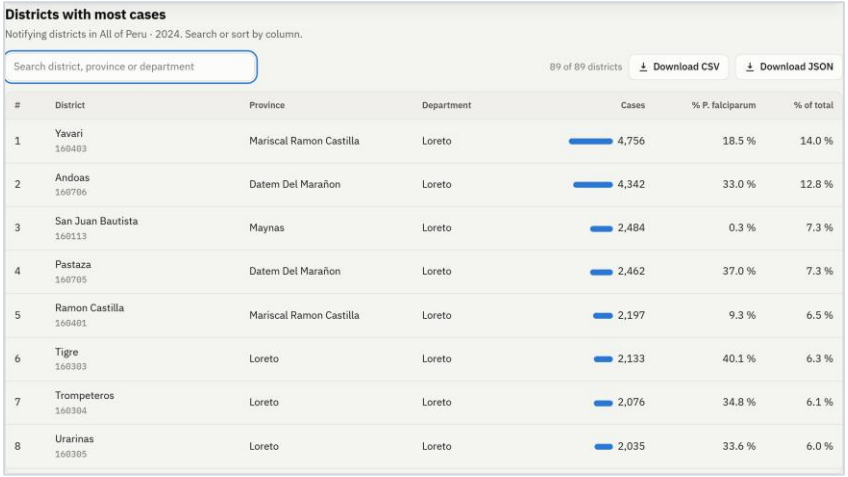
The web dashboard (2/2): surveillance views

Endemic channel · territory · population · districts with export









Point 1 — Adopted characteristics and priorities

All 9 ISO/IEC 25010:2023 characteristics adopted · 24 sub-characteristics selected · 14 justified as N/A or not evaluated

Characteristic	Priority	Selected sub-characteristics	Metrics
Functional suitability	High	Functional completeness, Functional correctness, Functional appropriateness	IF-01, IF-02, IF-03
Performance efficiency	Medium	Time behaviour, Resource utilization	ED-01, ED-02, ED-03
Compatibility	Low	Interoperability	CO-01
Interaction capability	High	Operability, User error protection, Inclusivity, Self-descriptiveness	CI-01, CI-02, CI-03, CI-04
Reliability	High	Faultlessness, Availability, Fault tolerance, Recoverability	FI-01, FI-02, FI-03, FI-04
Security	High	Confidentiality, Integrity, Resistance	SE-01, SE-02, SE-03
Maintainability	Medium	Analysability, Modifiability, Testability	MA-01, MA-02, MA-03
Flexibility	Medium	Adaptability	FL-01
Safety	High	Hazard warning, Fail safe, Risk identification	SO-01, SO-02, SO-03

Point 1 — What we did not evaluate, and why

Exclusions are explicit: not applicable (the property does not exist) vs. not evaluated (needs resources out of scope)

Characteristic	Sub-characteristic	Status	Justification
Performance efficiency	Capacity	Not applicable	Static site served by Vercel's CDN; no own server or concurrent users to limit.
Compatibility	Co-existence	Not applicable	Runs in the browser without sharing resources with other installed products.
Interaction capability	Appropriateness recognizability	Not evaluated	Requires tests with real CDC users; left for a user-testing iteration.
Interaction capability	Learnability	Not evaluated	Requires user testing; outside the scope of this automated evaluation.
Security	Non-repudiation	Not applicable	Read-only system: there are no transactions or user actions to attribute.
Security	Accountability	Not applicable	There are no user accounts or operations to log.
Security	Authenticity	Not applicable	Public information without sign-in.
Maintainability	Modularity	Not evaluated	Observed indirectly through MA-01 and MA-02; it has no metric of its own.
Maintainability	Reusability	Not evaluated	Not a project goal at this stage.
Flexibility	Scalability	Not applicable	Published data volume is fixed per version and hosting scales automatically.
Flexibility	Installability	Not applicable	Web application: nothing is installed.
Flexibility	Replaceability	Not applicable	Does not replace another product at this stage.
Safety	Operational constraint	Not applicable	The system does not control equipment or physical processes.
Safety	Safe integration	Not applicable	It is not integrated with other systems in this version.

Point 2 — Metrics defined with GQM

25 metrics · X = A/B × 100 or measured value · thresholds fixed BEFORE run 01

Goal → Question → Metric (example IF-02)

G: Ensure every figure shown matches the official CSV files.

Q: What share of the figures shown equals the value recomputed from the raw data?

M: X = A/B × 100 · A = matching values · B = compared values · threshold ≥ 100 %

Independent oracle

evaluacion/oraculo.py recomputes 8 scenarios (year × department × species) straight from the CSV files with its own INEI map.

Last run: 84 of 84 values identical.

Score range	Category (Calabrese et al., CACIC 2017)
0 – 40	Not acceptable
40 – 60	Minimally acceptable
60 – 90	Target range
90 – 100	Exceeds requirements

- **Proportion metrics:** score = X.
- **Lower-is-better values:** score = min(100, 100 × threshold / value).
- **Status:** Approved / Rejected against the threshold; Pending if not measured.
- **Owners:** every metric has a named team member.

Point 2 — Metrics owned by Daniel Wilston Chura Monroy

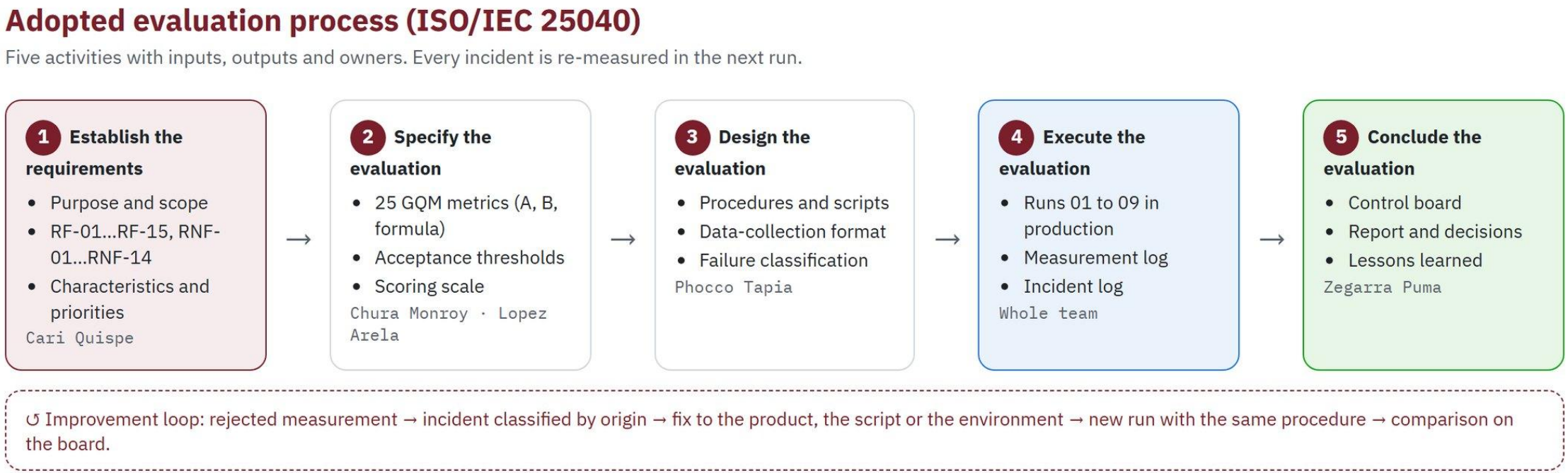
ID	Sub-characteristic	Metric	A / B (or value)	Threshold	Tool
IF-01	Functional completeness	Functional requirement coverage	A/B × 100	≥ 95 %	Playwright 1.64 (Chromium)
IF-02	Functional correctness	Accuracy against the official oracle	A/B × 100	≥ 100 %	Python 3.12 + Playwright
IF-03	Functional appropriateness	Coverage of surveillance dimensions	A/B × 100	≥ 90 %	Playwright
ED-01	Time behaviour	Largest Contentful Paint (mobile)	measured value	≤ 2.5 s	Lighthouse 12.8
ED-02	Time behaviour	Filter response time	measured value	≤ 200 ms	Playwright + Performance API
ED-03	Resource utilization	Initial page weight	measured value	≤ 1,000 KB	Lighthouse 12.8
CO-01	Interoperability	Valid and consistent exports	A/B × 100	≥ 100 %	Playwright
MA-01	Analysability	Functions with acceptable complexity	A/B × 100	≥ 90 %	ESLint 9 (regla complexity) + Radon 6
MA-02	Modifiability	Static-analysis issue density	measured value	≤ 2 issues/KLOC	ESLint 9
MA-03	Testability	Requirements with automated tests	A/B × 100	≥ 90 %	Script de consolidación
FL-01	Adaptability	Correct browser × viewport combinations	A/B × 100	≥ 100 %	Playwright

Point 2 — Metrics owned by Ower Frank Lopez Arela

ID	Sub-characteristic	Metric	A / B (or value)	Threshold	Tool
CI-01	Operability	Keyboard-operable controls	A/B × 100	≥ 95 %	Playwright
CI-02	User error protection	Invalid inputs handled	A/B × 100	≥ 100 %	Playwright
CI-03	Inclusivity	Serious WCAG violations	A/B × 100	≥ 100 %	axe-core 4 + Playwright
CI-04	Self-descriptiveness	Charts with a text alternative	A/B × 100	≥ 100 %	Playwright
FI-01	Faultlessness	Runs without JavaScript errors	A/B × 100	≥ 100 %	Playwright (Chromium, Firefox, WebKit)
FI-02	Availability	Measured availability	A/B × 100	≥ 99 %	Node.js fetch
FI-03	Fault tolerance	Data failures handled	A/B × 100	≥ 100 %	Playwright (page.route)
FI-04	Recoverability	Recovery after failure	A/B × 100	≥ 100 %	Playwright
SE-01	Confidentiality	No published microdata	A/B × 100	≥ 100 %	Node.js
SE-02	Integrity	Source data integrity	A/B × 100	≥ 100 %	Node.js crypto
SE-03	Resistance	OWASP security headers	A/B × 100	≥ 85 %	Node.js fetch
SO-01	Hazard warning	Views with a provisional-data warning	A/B × 100	≥ 100 %	Playwright
SO-02	Fail safe	No partial figures on failure	A/B × 100	≥ 100 %	Playwright
SO-03	Risk identification	Source limitations disclosed	A/B × 100	≥ 100 %	Playwright

Point 3 — Evaluation process (ISO/IEC 25040)

Five activities with inputs, outputs and owners; every rejected metric loops back through an incident



Point 3 — Data-collection format

One record per metric and run, seven blocks, generated from the evidence and mirrored in Excel

Data-collection format: one record per metric and run

The seven blocks are filled automatically from the evidence (registro.json) and mirrored in the Excel “Registro” sheet.

1. Identification

- Metric ID
- Characteristic
- Sub-characteristic

2. Traceability

- Requirement (RF/RNF)
- Test case or procedure
- Script or tool

3. Conditions

- Run and version
- Date and URL
- Environment and browser
- Data (SHA-256)

4. Measurement

- A and B (or value)
- Formula
- Result X and unit

5. Evaluation

- Threshold and expected result
- Status (approved, rejected, pending)
- Score and category

6. Evidence

- Run JSON file
- Screenshots and traces
- HTML report

7. Follow-up

- Owner
- Linked incidents
- Decision and notes

Possible statuses

- **Approved**: meets the threshold
- **Rejected**: does not meet it
- **Pending**: not measured
- **Not applicable**: justified sub-characteristic

Point 3 — Classifying every failure by its origin

22 incidents: 17 product · 4 test script · 1 infrastructure

Classifying a failure by its origin

Before fixing, every failure is reproduced and classified, so the product is not “fixed” when the error was in the test, or the other way round.

A test or measurement does not meet expectations

Product

The defect reproduces by hand on the published version (figures, UI, accessibility, performance, code).

- INC-01, 02, 11, 16, 18, 19, 21, 22 interface
- INC-07, 09, 17, 20 code and data
- INC-08, 12 to 15 performance

Action: fix the product, publish and re-measure.

Test script

The product behaves correctly; the failure is in the locator, the assertion or how the measurement is recorded.

- INC-04 ambiguous locator
- INC-05 lost rows
- INC-06, INC-10 false positives

Action: fix the script without touching the product.

Infrastructure

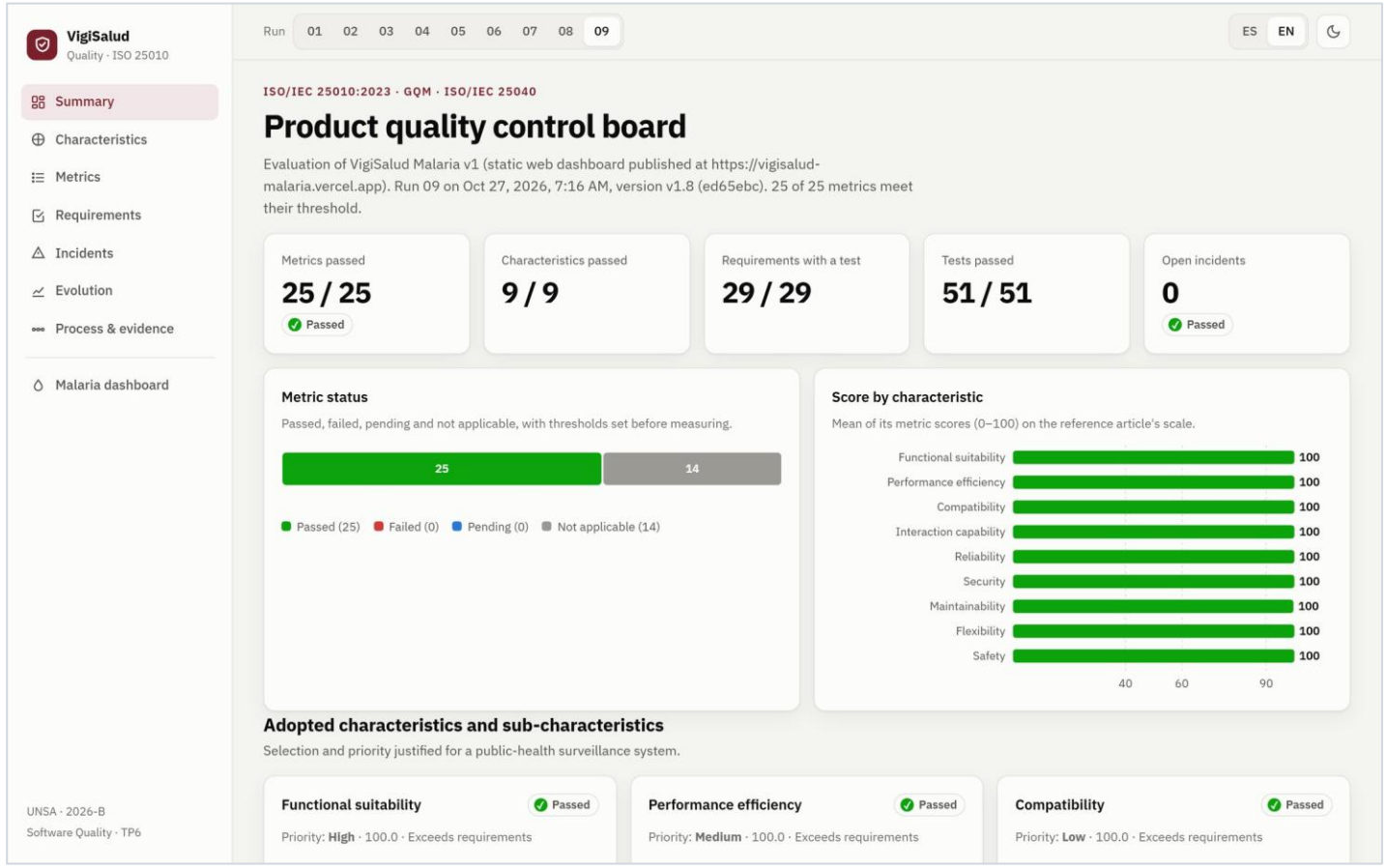
The failure depends on the environment: browsers, network, hosting or tooling.

- INC-03 browsers not installed

Action: repair the environment and repeat the run with the same procedure.

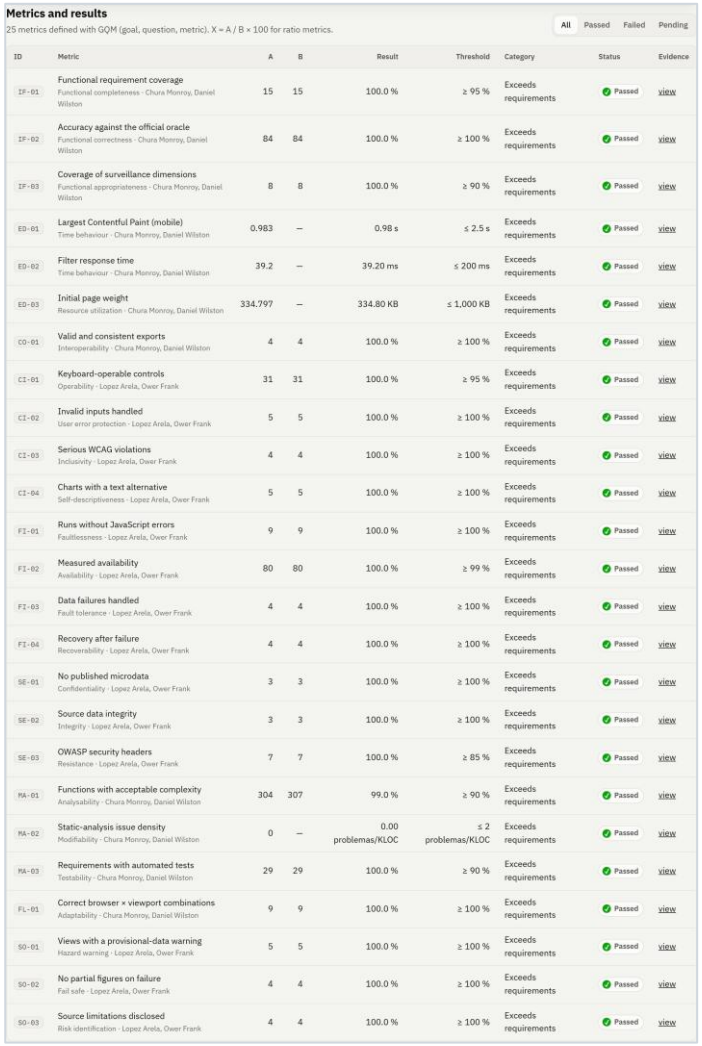
Point 4 — The control board (what it shows and how to read it)

https://vigisalud-malaria.vercel.app/calidad.html · bilingual · regenerated from the evidence after every run



- **Status:** approved / rejected / pending / N/A
- **Score per characteristic:** Calabrese scale
- **Metric detail:** A, B, X, threshold, evidence link
- **Requirement coverage:** 15 RF + 14 RNF
- **Incidents:** by origin and state
- **Evolution:** runs 01 → 09
- **Evidence:** screenshots of the last run

Point 4 — Board detail: metrics and incidents

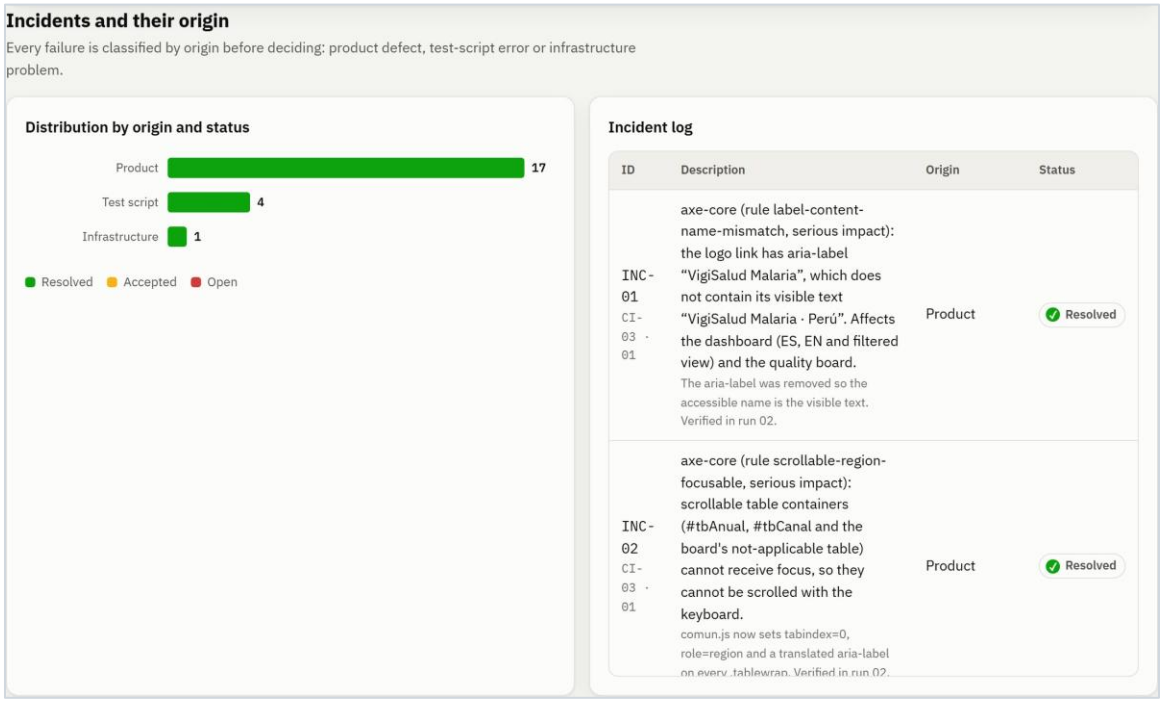


Metrics and results

25 metrics defined with GQM (goal, question, metric). X = A / B × 100 for ratio metrics.

All | Passed | Failed | Pending

ID	Metric	A	B	Result	Threshold	Category	Status	Evidence
IF-01	Functional requirement coverage Functional completeness · Chura Monroy, Daniel Wilston	15	15	100.0 %	≥ 95 %	Exceeds requirements	Passed	view
IF-02	Accuracy against the official oracle Functional correctness · Chura Monroy, Daniel Wilston	84	84	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
IF-03	Coverage of surveillance dimensions Functional appropriateness · Chura Monroy, Daniel Wilston	8	8	100.0 %	≥ 90 %	Exceeds requirements	Passed	view
ED-01	Largest Contentful Paint (mobile) Time behaviour · Chura Monroy, Daniel Wilston	0.983	–	0.98 s	≤ 2.5 s	Exceeds requirements	Passed	view
ED-02	Filter response time Time behaviour · Chura Monroy, Daniel Wilston	39.2	–	39.20 ms	≤ 200 ms	Exceeds requirements	Passed	view
ED-03	Initial page weight Resource utilization · Chura Monroy, Daniel Wilston	334.797	–	334.80 KB	≤ 1,000 KB	Exceeds requirements	Passed	view
CO-01	Valid and consistent exports Interoperability · Chura Monroy, Daniel Wilston	4	4	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
CI-01	Keyboard-operable controls Operability · Lopez Arela, Ower Frank	31	31	100.0 %	≥ 95 %	Exceeds requirements	Passed	view
CI-02	Invalid inputs handled User error protection · Lopez Arela, Ower Frank	5	5	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
CI-03	Serious WCAG violations Inclusivity · Lopez Arela, Ower Frank	4	4	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
CI-04	Charts with a text alternative Self-descriptiveness · Lopez Arela, Ower Frank	5	5	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
FI-01	Runs without JavaScript errors Faultlessness · Lopez Arela, Ower Frank	9	9	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
FI-02	Measured availability Availability · Lopez Arela, Ower Frank	80	80	100.0 %	≥ 99 %	Exceeds requirements	Passed	view
FI-03	Data failures handled Fault tolerance · Lopez Arela, Ower Frank	4	4	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
FI-04	Recovery after failure Recoverability · Lopez Arela, Ower Frank	4	4	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
SE-01	No published microdata Confidentiality · Lopez Arela, Ower Frank	3	3	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
SE-02	Source data integrity Integrity · Lopez Arela, Ower Frank	3	3	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
SE-03	OWASP security headers Resistance · Lopez Arela, Ower Frank	7	7	100.0 %	≥ 85 %	Exceeds requirements	Passed	view
MA-01	Functions with acceptable complexity Analysability · Chura Monroy, Daniel Wilston	304	307	99.0 %	≥ 90 %	Exceeds requirements	Passed	view
MA-02	Static-analysis issue density Modifiability · Chura Monroy, Daniel Wilston	0	–	0.00 problemas/KLOC	≤ 2 problemas/KLOC	Exceeds requirements	Passed	view
MA-03	Requirements with automated tests Testability · Chura Monroy, Daniel Wilston	29	29	100.0 %	≥ 90 %	Exceeds requirements	Passed	view
FL-01	Correct browser × viewport combinations Adaptability · Chura Monroy, Daniel Wilston	9	9	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
SO-01	Views with a provisional-data warning Hazard warning · Lopez Arela, Ower Frank	5	5	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
SO-02	No partial figures on failure Fail safe · Lopez Arela, Ower Frank	4	4	100.0 %	≥ 100 %	Exceeds requirements	Passed	view
SO-03	Source limitations disclosed Risk identification · Lopez Arela, Ower Frank	4	4	100.0 %	≥ 100 %	Exceeds requirements	Passed	view



Incidents and their origin

Every failure is classified by origin before deciding: product defect, test-script error or infrastructure problem.

Distribution by origin and status

Incident log

ID	Description	Origin	Status
INC-01 CI-03 · 01	axe-core (rule label-content-name-mismatch, serious impact): the logo link has aria-label “VigiSalud Malaria”, which does not contain its visible text “VigiSalud Malaria · Perú”. Affects the dashboard (ES, EN and filtered view) and the quality board. The aria-label was removed so the accessible name is the visible text. Verified in run 02.	Product	Resolved
INC-02 CI-03 · 01	axe-core (rule scrollable-region-focusable, serious impact): scrollable table containers (#tbAnual, #tbCanal and the board's not-applicable table) cannot receive focus, so they cannot be scrolled with the keyboard. comun.js now sets tabindex=0, role=region and a translated aria-label on every .tablewrap. Verified in run 02.	Product	Resolved

Evolution on the control board

The board keeps every run: status per metric, conditions and the version evaluated

Evolution across runs

Each run executes the same tests on the published version; fixes between runs respond to logged incidents.

Metrics by status in each run



Conditions of each run

Run	Date	Version	Tests
01	10/27/26, 12:09 AM Windows_NT 10.0.26300 · Node v22.14.0 · Playwright Version 1.64.1 · Lighthouse 12.8.2	v1.0 (e2ec6c9)	40/51
02	10/27/26, 12:22 AM Windows_NT 10.0.26300 · Node v22.14.0 · Playwright Version 1.64.1 · Lighthouse 12.8.2	v1.1 (a63333f)	51/51
03	10/27/26, 6:13 AM Windows_NT 10.0.26300 · Node v22.14.0 · Playwright Version 1.64.1 · Lighthouse 12.8.2	v1.2 (ecfaca3)	51/51
04	10/27/26, 6:22 AM Windows_NT 10.0.26300 · Node v22.14.0 · Playwright Version 1.64.1 · Lighthouse 12.8.2	v1.3 (1219f7a)	51/51
05	10/27/26, 6:34 AM Windows_NT 10.0.26300 · Node v22.14.0 · Playwright Version 1.64.1 · Lighthouse 12.8.2	v1.4 (ecd9c46)	51/51
06	10/27/26, 6:44 AM Windows_NT 10.0.26300 · Node v22.14.0 · Playwright Version 1.64.1 · Lighthouse 12.8.2	v1.5 (2aca5a2)	51/51

Results across the runs

Approved metrics per run: 18 → 24 → 24 → 24 → 25 → 25 → 25 → 25 → 25 (of 25) · automated tests: 40/51 → 51/51

ID	Metric	Threshold	01	02	03	04	05	06	07	08	09
IF-01	Functional requirement coverage	≥ 95 %	93.3 ✗	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
IF-02	Accuracy against the official oracle	≥ 100 %	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
ED-01	Largest Contentful Paint (mobile)	≤ 2.5 s	3.29 ✗	2.48	2.56 ✗	2.55 ✗	0.98	0.98	0.98	0.98	0.98
ED-02	Filter response time	≤ 200 ms	42.80	42.40	38.50	38.50	36.90	36.60	37.10	38.80	39.20
CI-03	Serious WCAG violations	≥ 100 %	0.0 ✗	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
FI-01	Runs without JavaScript errors	≥ 100 %	33.3 ✗	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
FI-02	Measured availability	≥ 99 %	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
SE-01	No published microdata	≥ 100 %	66.7 ✗	66.7 ✗	100.0	100.0	100.0	100.0	100.0	100.0	100.0
MA-02	Static-analysis issue density	≤ 2 issues/KLOC	3.01 ✗	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
FL-01	Correct browser × viewport combinations	≥ 100 %	33.3 ✗	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Columns 01–09: result X per run (units as in the threshold). Rows: metrics rejected in at least one run, plus three reference metrics. ✗ = threshold not met.

Incident log (1/2)

22 incidents: 17 product · 4 test script · 1 infrastructure — all reproduced, classified, fixed and re-measured

ID	Run	Metric	Origin	What happened
INC-01	01	CI-03	Product	axe-core (rule label-content-name-mismatch, serious impact): the logo link has aria-label “VigiSalud Malaria”, which does not contain its…
INC-02	01	CI-03	Product	axe-core (rule scrollable-region-focusable, serious impact): scrollable table containers (#tbAnual, #tbCanal and the board's not-applicable…
INC-03	01	FL-01	Infrastructure	The 6 Firefox and WebKit combinations (mobile, tablet and desktop) did not run: “browserType.launch: Executable doesn't exist” on the test…
INC-04	01	IF-01	Script	Test RF-04 failed with a “strict mode violation”: the locator getByRole('button', {name: 'P. falciparum'}) matched both the species filter…
INC-05	01	CI-03	Script	ci03-axe.json kept only the last of the 4 audited views. When a test fails Playwright restarts the worker, and rows held in memory were…
INC-06	01	SE-01	Script	SE-01 flagged calidad_datos.json as individual data because it contains the key “tipo_edad”. On review, that key holds an aggregate count…
INC-07	01	MA-02	Product	ESLint reported 3 no-unused-vars problems (unused “e” in catch blocks of comun.js and calidad.js): 3.01 per KLOC, above the threshold of 2.
INC-08	01	ED-01	Product	Mobile LCP of 3.29 s (median of 3 Lighthouse runs), above 2.5 s. The LCP element is the provisional-data notice; 79 % of the time is render…
INC-09	01	MA-01	Product	Function kpis() in app.js had cyclomatic complexity 25, more than twice the limit of 10, although MA-01 overall (98.2 %) met the threshold.
INC-10	02	SE-01	Script	SE-01 flagged calidad_datos.json again, this time for “localidad” and “localcod” inside “nulos” ({localidad: 5412, localcod: 5634}), which…
INC-11	02	CI-03	Product	Lighthouse's mobile accessibility audit dropped from 100 to 96 (rule link-name): below 960 px the logo link had no accessible name. This is…

Incident log (2/2)

22 incidents: 17 product · 4 test script · 1 infrastructure — all reproduced, classified, fixed and re-measured

ID	Run	Metric	Origin	What happened
INC-12	02	ED-01	Product	Cumulative layout shift (CLS) rose from 0 to 0.11 on mobile, above the Core Web Vitals “good” value (0.1). Lighthouse attributes it to the…
INC-13	03	ED-01	Product	Mobile CLS stayed at 0.109 after INC-12. PerformanceObserver (layout-shift) at 412×823 showed the real cause: when the data is painted, the…
INC-14	03	ED-01	Product	Mobile LCP of 2.56 s (median of 3 runs), above the 2.5 s threshold; it was 2.48 s in run 02. The LCP element (the provisional-data notice)…
INC-15	04	ED-01	Product	CLS was fixed (0.0013), but mobile LCP stayed at 2.55 s even though the notice no longer waited for the data. Lighthouse uses simulated…
INC-16	05	CI-04	Product	A manual review of the run 05 screenshots showed the annual chart legend displaying “P.” and “…lcipar” instead of the species names. The…
INC-17	06	IF-03	Product	A manual review of the run 06 screenshots showed “Datem Del Mara□on” in the districts table. The official CSV files have the letter Ñ…
INC-18	06	CI-04	Product	In the endemic channel, the “Alarm” and “Safety” labels overlapped at the end of the chart when those bands are thin (e.g. 2024).
INC-19	06	IF-03	Product	In the “Data validation” card, the long label “Department recovered from INEI ubigeo” widened the first column and left values in a very…
INC-20	06	SE-02	Product	Re-running the pipeline for INC-17 revealed it took the download date from the system date (date.today()): each run would have changed the…
INC-21	07	IF-01	Product	Reviewing the English screenshots of run 07 showed Spanish text in the English interface. An automated sweep of the visible text confirmed…
INC-22	08	IF-01	Product	Visual inspection of the English screenshots of run 08 showed “Priority: Alta / Media / Baja” in the board's characteristic cards. The…

Lessons learned

- **Classify before fixing:** 5 of 22 incidents were in our scripts or environment, not in the product.
- **Re-measure after every fix:** two fixes caused regressions (INC-11, INC-12) and one exposed a latent layout defect (INC-13).
- **Close incidents with evidence:** INC-12 looked fixed from the tool's explanation; run 03 proved it was not.
- **One optimisation can hurt another:** preloads that helped LCP in run 02 hurt it once the LCP element no longer needed the data (INC-15).
- **Green is not the end:** visual review found a clipped legend, overlapping labels, Ñ damaged in the official files and Spanish text in the English UI (INC-16 to INC-22); visual and automated checks complement each other.
- **Respect the source:** restored characters are listed and every figure was checked to be unchanged.

Conclusions

9 / 9
characteristics adopted

25
GQM metrics, automated

18 → 25
approved metrics (of 25)

22
incidents classified & closed

- **Point 1:** explicit priorities and justified exclusions for a read-only public-health system.
- **Point 2:** every selected sub-characteristic has a metric with formula, source, owner and a threshold fixed before measuring.
- **Point 3:** ISO/IEC 25040 process, seven-block collection format, failures classified by origin before fixing.
- **Point 4:** public bilingual control board regenerated from the evidence of every run.
- **Next steps:** run the evaluation in CI with deployment gates; add mobile axe audit; user testing for learnability.

Dashboard: https://vigisalud-malaria.vercel.app

Quality board: https://vigisalud-malaria.vercel.app/calidad.html · **Repository:** https://github.com/OwerLopez/vigisalud-malaria-iso25010